CHAPTER 8: LECTURE 1

Def'n Let $m \in \mathbb{N}$ and $a, b \in \mathbb{Z}$. If $m \mid (a-b)$, we say <u>a is congruent to b modulo m</u>, denoted by $a \equiv b \pmod{m}$.

If $m \nmid (a-b)$, then we write $a \not\equiv b \pmod{m}$

EX $9 \equiv 1 \pmod{4}$ since $4 \mid (9-1)$ but $9 \not\equiv 1 \pmod{3}$ since $3 \nmid (9-1)$.

$-3 \equiv 7 \pmod{5}$ since $5 \mid (-3-5)$.

$a \equiv b \pmod{2}$ if a and b are both odd or even

Remark $a \equiv 0 \pmod{m}$ iff $m \mid a$

Proposition <u>Congruence is an Equivalent Relation (CER)</u>

Let $a, b, c \in \mathbb{Z}$ and $m \in \mathbb{N}$. We have

(1) $a \equiv a \pmod{m}$ Reflectivity

(2) If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$ Symmetry

(3) If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$. Transitivity

Def'n: $m \mid (a-b) \Rightarrow a \equiv b \pmod{m}$
$m \mid a \Rightarrow a \equiv 0 \pmod{m}$

Proposition: CER

CHAPTER 8: LECTURE 2

Remark

$a \equiv b \pmod{m} \iff m \mid (a-b)$
$\iff \exists\, k \in \mathbb{Z}$ such that $a - b = km$
$\iff \exists\, k \in \mathbb{Z}$ such that $a = km + b$

In particular, if $a \equiv b \pmod{m}$, then

$$a \in \{\ldots, b-2m, b-m, b, b+m, b+2m, \ldots\}$$

EX

If $x \equiv 1 \pmod{4}$, then

$$x \in \{\ldots, -7, -3, 1, 5, 9, \ldots\}$$

If $x \equiv 1 \pmod{2}$, then

$$x \in \{\ldots, -3, -1, 1, 3, 5, \ldots\}$$

Proposition

<u>Arithmetic Rules</u>

Let $m \in \mathbb{N}$ and $a_1, a_2, b_1, b_2 \in \mathbb{Z}$ If $a_1 \equiv b_1 \pmod{m}$ and $a_2 \equiv b_2 \pmod{m}$, then

(1) $a_1 + a_2 \equiv b_1 + b_2 \pmod{m}$

(2) $a_1 - a_2 \equiv b_1 - b_2 \pmod{m}$

(3) $a_1 a_2 \equiv b_1 b_2 \pmod{m}$

EX

$9 \equiv 1 \pmod{4}$ and $7 \equiv 3 \pmod{4}$. Thus we have

(1) $9 + 7 \equiv 1 + 3 \pmod{4}$, i.e. $16 \equiv 4 \pmod{4}$

(2) $9 - 7 \equiv 1 - 3 \pmod{4}$, i.e. $2 \equiv -2 \pmod{4}$

(3) $9 \cdot 7 \equiv 1\ 3 \pmod{4}$, i.e. $63 \equiv 3 \pmod{4}$

by Arithmetic Rules of Congruency

Remark: $m \mid (a-b) \iff \exists\, k \in \mathbb{Z}$ such that $a - b = km$

Proposition: Congruency addition, subtraction, multiplication

CONTINUED

Proposition <u>Congruence Add and Multiply (CAM)</u>

Let $m, n \in \mathbb{Z}$. Suppose that $a_i \equiv b_i \pmod{m}$ with $1 \leq i \leq n$. Then, we have

(1) $a_1 + a_2 + \dots + a_n \equiv b_1 + b_2 + \dots + b_n \pmod{m}$

(2) $a_1 a_2 \dots a_n \equiv b_1 b_2 \dots b_n \pmod{m}$

Proposition <u>Congruence Power (CP)</u>

For $m, n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$, if $a \equiv b \pmod{m}$, then $a^n \equiv b^n \pmod{m}$

Proposition <u>Congruence Divide (CD)</u>

For $m \in \mathbb{N}$ and $a, b, c \in \mathbb{Z}$, if $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proposition <u>Congruence iff Same Remainder (CISR)</u>

For $m \in \mathbb{N}$ and $a, b \in \mathbb{Z}$, $a \equiv b \pmod{m}$ iff a and b have the same remainder when divided by m.

Proposition <u>Congruence to Remainder (CTR)</u>

For $m \in \mathbb{N}$ and $a, b \in \mathbb{Z}$ with $0 \leq b < m$, $a \equiv b \pmod{m}$ iff a has the remainder b when divided by m.

Ex Determine the remainder when $33^{49} + 63^{50}$ is divided by 32.

Consider mod 32 We have $33 \equiv 1 \pmod{32}$ and $63 \equiv 31 \equiv -1 \pmod{32}$. Thus we have

$$33^{49} + 63^{50} \equiv 1^{49} + (-1)^{50} \equiv 1 + 1 \equiv 2 \pmod{32}$$

Thus, the remainder is 2.

Propositions: CAM, CP, CTR, CISR, CD

CONTINUED

Ex

Determine the last decimal digits of 113^{55} and 114^{55}

Note that the last decimal digit is the remainder when divided by 10

Consider mod 10. We have

$113 \equiv 3 \pmod{10}$ Thus, $113^{55} \equiv 3^{55} \pmod{10}$

Note that $3^2 \equiv 9 \equiv -1 \pmod{10}$ Thus we have

$$\begin{aligned} 113^{55} &\equiv 3^{55} \\ &\equiv (3^2)^{27} \cdot 3 \\ &\equiv (-1)^{27} \cdot 3 \\ &\equiv -1 \cdot 3 \\ &\equiv -3 \\ &\equiv 7 \pmod{10} \end{aligned}$$

Hence the last decimal digit of 113^{55} is 7.

We have $114 \equiv 4 \pmod{10}$. Thus $114^{55} \equiv 4^{55} \pmod{10}$.

Note that $4^3 \equiv 64 \equiv 4 \pmod{10}$. Thus we have

$$\begin{aligned} 114^{55} &\equiv 4^{55} \\ &\equiv (4^3)^{18} \quad 4 \\ &\equiv 4^{18} \quad 4 \\ &\equiv (4^3)^6 \cdot 4 \\ &\equiv 4^6 \cdot 4 \\ &\equiv (4^3)^2 \cdot 4 \\ &\equiv 4^2 \cdot 4 \\ &\equiv 16 \cdot 4 \\ &\equiv 64 \\ &\equiv 4 \pmod{10} \end{aligned}$$

Hence the last decimal digit of 114^{55} is 4.

Choose efficient powers to work with

CHAPTER 8: LECTURE 3

Proposition A number is divisible by 3 iff the sum of its digits is divisible by 3.

The same result holds if we replace 3 with 9 since $10 \mid 1 \pmod 9$.

Proposition A number is divisible by 11 iff the alternating sum of its digits is divisible by 11.

Ex Let $n = 555 \cdots 5$ be a 200-digit number. Is n divisible by 3 or 11?

Note that the sum of digits $= 5 \cdot 200 = 1000$ and the alternating sum $= (5-5) \cdot 100 = 0$

Since $3 \nmid 1000$ and $0 \mid 11$, n is divisible by 11 but not 3.

Def'n Let $m \in \mathbb{N}$ and $a, c \in \mathbb{Z}$. A relation of the form $ax \equiv c \pmod m$ is called a <u>linear congruence</u> in the variable x.

Proposition The linear congruence $ax \equiv c \pmod m$ has a solution $x = x_0$ iff the linear diophantine equation $ax + my = c$ has a solution $x = x_0$, $y = y_0$ for some $y = y_0 \in \mathbb{Z}$.

Ex Find $x \in \mathbb{Z}$ such that $7x \equiv 9 \pmod{38}$.

It is equivalent to solve $7x + 38y = 9$. By EEA,

x	y	r	q
0	1	38	0
1	0	7	0
-5	1	3	5
11	-2	1	2
-38	7	0	3

Thus, $\gcd(7, 38) = 1$ and $1 = 7 \cdot 11 + 38 \cdot -2$.

Since $9 = 9 \cdot 1$,
$9 = 7(99) + 38(-18)$.

Thus $x = 99$ is a solution.

Propositions: divisibility by 3, 9, 11

Def'n: Linear Congruence

Proposition: Solving linear congruencies through linear diophantine equations

CONTINUED

Proposition <u>Linear Congruence Theorem (LCT)</u>

Let $a, c \in \mathbb{Z}$ with $a \neq 0$.

(1) $ax \equiv c \pmod{m}$ has a solution iff $\gcd(a, m) \mid c$.

(2) If $x = x_0$ is a solution of $ax \equiv c \pmod{m}$, then the complete solution is $x \equiv x_0 \pmod{\frac{m}{d}}$ where $d = \gcd(a, m)$.

Remark Note that $x \equiv x_0 \pmod{\frac{m}{d}}$ is equivalent to

$$x \equiv x_0,\ x_0 + \frac{m}{d},\ x_0 + 2\frac{m}{d},\ \dots,\ x_0 + (d-1)\frac{m}{d} \pmod{m}$$

Note that $x_0 + d\frac{m}{d} \equiv x_0 + m \equiv x_0 \pmod{m}$

For example, $x \equiv 1 \pmod{2}$ is equivalent to $x \equiv 1, 3 \pmod{4}$ and $x \equiv 1, 3, 5, 7 \pmod{8}$

Ex Find, if possible, the complete sol'n to $9x \equiv 7 \pmod{15}$

Note that $\gcd(9, 15) = 3$. Since $3 \nmid 7$, by LCT, there is no solution.

Ex Find, if possible, the complete sol'n to $35x \equiv 10 \pmod{125}$.

It is equivalent to solving $35x + 125y = 10$. By EEA,

x	y	r	q
0	1	125	0
1	0	35	0
-3	1	20	3
4	-1	15	1
-7	2	5	1
25	-7	0	3

Thus, $\gcd(35, 125) = 5$ and $35 \cdot -7 + 125 \cdot 2 = 5$.

Since $10 = 5 \cdot 2$, $10 = 35 \cdot -14 + 125 \cdot 4$

Thus $x = -14$ is a solution.

Hence the complete solution is $x \equiv -14 \equiv 11 \pmod{25}$, i.e. $x \equiv 11, 36, 61, 86, 111 \pmod{125}$.

Proposition: LCT (similar to LDET2)

CHAPTER 8. LECTURE 4

Def'n

Congruence Class and Modular Arithmetic

Consider $\ldots, -6, -3, 0, 3, 6, 9, \ldots,$ which are all $\equiv 0 \pmod 3$.

We use the notation $[0] = \{ n \in \mathbb{Z} : n \equiv 0 \pmod 3 \}$
$= \{ \ldots, -6, -3, 0, 3, 6, \ldots \}$

Similarly $[1] = \{ n \in \mathbb{Z} : n \equiv 1 \pmod 3 \}$
$= \{ \ldots, -5, -2, 1, 4, 7, \ldots \}$

and $[2] = \{ n \in \mathbb{Z} : n \equiv 2 \pmod 3 \}$
$= \{ \ldots, -4, -1, 2, 5, 8, \ldots \}$

Note that $[3] = \{ n \in \mathbb{Z} : n \equiv 3 \pmod 3 \}$
$= \{ \ldots, -3, 0, 3, 6, 9, \ldots \}$
$= [0]$

Indeed $[-3] = [0] = [3] = [6] = \ldots$

Similarly, $\ldots = [-2] = [1] = [4] = [7] = \ldots$
$\ldots = [-1] = [2] = [5] = [8] = \ldots$

Thus it is enough to consider only $[0], [1], [2]$, where $0, 1, 2$ are all possible remainders when we divide by 3.

Since $[0], [1], [2]$ contain all integers, we have

$$\mathbb{Z}_3 = \{ [0], [1], [2] \}$$

new way to write modulos using $[x]$

CONTINUED

Def'n Let $m \in \mathbb{N}$. For $k \in \mathbb{Z}$, we define

$$[k] = \{ n \in \mathbb{Z} : n \equiv k \pmod{m} \}$$
$$= \{ \ldots, k-2m, k-m, k, k+m, k+2m, \ldots \}$$

We call k a <u>congruence class of m</u>. Note that for any $n \in \mathbb{Z}$, $n \equiv k \pmod{m}$ for exactly one of $k = 0, 1, 2, \ldots, (m-1)$.

Thus n is contained in exactly one of $[0], [1], \ldots, [m-1]$. Since these sets contain all integers, we denote $\mathbb{Z}_m$ as their union, i.e. $\mathbb{Z}_m = \{ [0], [1], [2], \ldots, [m-1] \}$.

We call $\mathbb{Z}_m$ the set of <u>integers modulo m</u>.

Remark In general, $[a] = [b]$ in $\mathbb{Z}_m \Leftrightarrow a \equiv b \pmod{m}$.

Arithmetic calculation can be performed in $\mathbb{Z}_m$. In particular, $[a] + [b] = [a+b]$, $[a][b] = [ab]$.

Ex Write down the addition and multiplication tables of $\mathbb{Z}_5$.

$+$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$
$[0]$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$
$[1]$	$[1]$	$[2]$	$[3]$	$[4]$	$[0]$
$[2]$	$[2]$	$[3]$	$[4]$	$[0]$	$[1]$
$[3]$	$[3]$	$[4]$	$[0]$	$[1]$	$[2]$
$[4]$	$[4]$	$[0]$	$[1]$	$[2]$	$[3]$

$\cdot$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$
$[0]$	$[0]$	$[0]$	$[0]$	$[0]$	$[0]$
$[1]$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$
$[2]$	$[0]$	$[2]$	$[4]$	$[1]$	$[3]$
$[3]$	$[0]$	$[3]$	$[1]$	$[4]$	$[2]$
$[4]$	$[0]$	$[4]$	$[3]$	$[2]$	$[1]$

Remark We have $[a][x] = [c]$ in $\mathbb{Z}_m \Leftrightarrow ax \equiv c \pmod{m}$
$\Leftrightarrow ax + my = c$

Def'n: congruence class, modular arithmetic

CONTINUED

Proposition <u>Modular Arithmetic Theorem (MAT)</u>

Let $m \in \mathbb{N}$.

(1) The equation $[a][x] = [c]$ in $\mathbb{Z}_m$ has a solution iff $\gcd(a, m) \mid c$.

(2) If $[x] = [x_0]$ is a solution of $[a][x] = [c]$, then the complete solution is

$$[x_0], [x_0 + \tfrac{m}{d}], [x_0 + 2\tfrac{m}{d}], \ldots, [x_0 + (d-1)\tfrac{m}{d}]$$

where $d = \gcd(a, m)$. Note that there are d many solutions.

Ex Solve $[2][x] = [4]$ in $\mathbb{Z}_{16}$.

It is equivalent to solve $2x \equiv 4 \pmod{16}$, i.e. $2x + 16y = 4$.

x	y	r	q
0	1	16	0
1	0	2	0
-8	1	0	8

Thus, $\gcd(2, 16) = 2$ and $2 \cdot 1 + 16 \cdot 0 = 2$.

Since $4 = 2 \cdot 2$, $4 = 2 \cdot 2 + 0 \cdot 2$

Hence $x = 2$ is a solution

Thus, $[x] = [2]$ or $[2 + \frac{16}{2}] = [10]$ in $\mathbb{Z}_{16}$.

Def'n For $[a], [b] \in \mathbb{Z}_m$, if $[a][b] = [1]$, $[b]$ is the <u>multiplicative inverse of $[a]$</u>, denoted by $[a]^{-1}$.

Ex In $\mathbb{Z}_5$, $[2][3] = [6] = [1]$. Thus $[2]^{-1} = [3]$ and $[3]^{-1} = [2]$ Also, $[1]^{-1} = [1]$.

Since $[4][4] = [16] = [1]$ in $\mathbb{Z}_5$, we have $[4]^{-1} = [4]$.

Proposition: MAT (similar to LDET2)

Def'n: multiplicative inverse

CONTINUED

Ex

Find the multiplicative inverse of $[6]$ in $\mathbb{Z}_8$.

$\cdot$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$	$[5]$	$[6]$	$[7]$
$[6]$	$[0]$	$[6]$	$[4]$	$[2]$	$[0]$	$[6]$	$[4]$	$[2]$

Thus $[6]$ has no multiplicative inverse in $\mathbb{Z}_8$.

Corollary

<u>Inverse in $\mathbb{Z}_m$ (INV $\mathbb{Z}_m$)</u>

Let $a \in \mathbb{Z}$.

(1) $[a]^{-1}$ exists in $\mathbb{Z}_m$ iff $\gcd(a, m) = 1$.

(2) If $[a]^{-1}$ exists, then it is unique

Corollary : multiplicative inverse existence

CHAPTER 8 : LECTURE 5

Corollary <u>Inverse in $\mathbb{Z}_p$ (INV $\mathbb{Z}_p$)</u>

Let p be a prime and $[a] \neq [0]$ in $\mathbb{Z}_p$. Then $[a]^{-1}$ exists and is unique.

Ex Find $[7]^{-1}$ in $\mathbb{Z}_{38}$.

Let $x = [7]^{-1} \in \mathbb{Z}_{38}$. We have $[7][x] = [1]$ in $\mathbb{Z}_{38}$, i.e. $7x \equiv 1 \pmod{38}$. It is equivalent to solving $7x + 38y = 1$.

Since $\gcd(7, 38) = 1$ and $1 = 7 \cdot 11 + 38 \cdot -2$, we have $[x] = [11]$.

Theorem <u>Fermat's Little Theorem (FlT)</u>

If p is a prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Ex Find $2^{405} \pmod{11}$.

Since 11 is a prime and $11 \nmid 2$, by FlT, $2^{10} \equiv 1 \pmod{11}$. Consider mod 11. Then,

$$2^{405} \equiv (2^{10})^{40} \; 2^5 \equiv 1 \cdot 2^5 \equiv 32 \equiv 10 \pmod{11}$$

Corollary For any integer a and prime p, we have $a^p \equiv a \pmod{p}$.

Remark If the explicit value of a is given and $p \nmid a$, we use FlT. If the value of a is unknown, then it is better to use the corollary.

Proposition : INV $\mathbb{Z}_p$

Theorem : FlT

Corollary : FlT but with unknown a value

CONTINUED

EX What is the remainder when $6^{81} + 17^{64}$ is divided by 7?

Consider mod 7. Since $6 \equiv -1 \pmod 7$,
$6^{81} \equiv (-1)^{81} \equiv -1 \equiv 6 \pmod 7$.

Since $17 \equiv 3 \pmod 7$ and $3 \nmid 7$, by FLT,
$3^6 \equiv 1 \pmod 7$. Thus

$$17^{64} \equiv 3^{64} \equiv (3^6)^{10} \cdot 3^4 \equiv 1 \cdot 81 \equiv 4 \pmod 7$$

Thus, $6^{81} + 17^{64} \equiv 6 + 4 \equiv 10 \equiv 3 \pmod 7$

Hence the remainder is 3.

EX Let p and q be distinct primes. Prove that for all integers a, if $a^{pq} \not\equiv a \pmod q$, then $a^p \not\equiv a \pmod q$.

By the contrapositive law, it suffices to prove that if $a^p \equiv a \pmod q$, then $a^{pq} \equiv a \pmod q$.

Suppose $a^p \equiv a \pmod q$ Consider mod q.
By the assumption and the corollary of FLT, we have

$$a^{pq} \equiv (a^p)^q \equiv a^q \equiv a \pmod q.$$

Remark: $a^p \equiv a \pmod p$

CHAPTER 8 : LECTURE 6

Theorem <u>Chinese Remainder Theorem (CRT)</u>

Let $m_1, m_2 \in \mathbb{N}$ with $\gcd(m_1, m_2) = 1$.
For any $a_1, a_2 \in \mathbb{Z}$, the simultaneous congruencies

$$x \equiv a_1 \pmod{m_1} \quad \text{——} \quad (1)$$
$$x \equiv a_2 \pmod{m_2} \quad \text{——} \quad (2)$$

has a solution. Moreover, if $x = x_0$ is a solution, then the complete solution is
$x \equiv x_0 \pmod{m_1 m_2}$

Ex Find $x \in \mathbb{Z}$ such that $x \equiv 5 \pmod{9}$ —— (1) and
$x \equiv 3 \pmod{7}$ —— (2)

Such a relation is called a <u>simultaneous congruency</u>.

Suppose x satisfies (1) Then $x = 5 + 9y$ for some $y \in \mathbb{Z}$. Put it into (2). We get

$$5 + 9y \equiv 3 \pmod{7}, \text{ i.e. } 9y \equiv -2 \pmod{7}$$

Hence we have $2y \equiv -2 \pmod{7}$ —— (3)

By inspection, $y = -1$ is a solution to (3)
Thus $x = 5 + 9(-1) = -4$ is a solution to (1) and (2).

Theorem: CRT

Def'n: simultaneous congruency

CONTINUED

Ex

Find the complete solution to $\begin{cases} x \equiv 2 \pmod{16} & \text{—— (1)} \\ x \equiv 5 \pmod{37} & \text{—— (2)} \end{cases}$

Since $\gcd(16, 37) = 1$, by CRT, there is a solution. If x satisfies (1), $x = 2 + 16y$, for some $y \in \mathbb{Z}$.

Put it into (2). We have $2 + 16y \equiv 5 \pmod{37}$,

i.e. $16y \equiv 3 \pmod{37}$ —— (3).

It is equivalent to solve $16y + 37z = 3$.

y	z	r	q
0	1	37	0
1	0	16	0
-2	1	5	2
7	-3	1	3
-37	16	0	5

Thus, $\gcd(16, 37) = 1$ and $16 \cdot 7 + 37 \cdot -3 = 1$

Since $3 = 1 \cdot 3$, $16 \cdot 21 + 37 - 9 = 3$.

Thus $y = 21$ is a solution. It follows that $x = 2 + 16 \cdot 21 = 338$ is a solution. By CRT, the complete solution is

$x \equiv 338 \pmod{16 \cdot 37}$, i.e. $x \equiv 338 \pmod{592}$.

Theorem

<u>Splitting Modulus Theorem (SMT)</u>

Let $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. Then, for any $x, a \in \mathbb{Z}$, we have

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv a \pmod{n} \end{cases}$$

iff $x \equiv a \pmod{mn}$.

Theorem: SMT (ability to split mods for large numbers)

CONTINUED

Ex Determine all solutions to the equation

$$[x]^{13} + [7][x] + [5] = [0] \text{ in } \mathbb{Z}_{91}$$

It is equivalent to solve $x^{13} + 7x + 5 \equiv 0 \pmod{91}$.

Since $91 = 7 \cdot 13$ and $\gcd(7, 13) = 1$, by SMT, it is equivalent to solve

$$\begin{cases} x^{13} + 7x + 5 \equiv 0 \pmod{7} \\ x^{13} + 7x + 5 \equiv 0 \pmod{13} \end{cases}$$

Consider mod 7. By the corollary of FlT, we have $x^7 \equiv x \pmod{7}$. Thus,

$$x^{13} \equiv x^7 \cdot x^6 \equiv x \cdot x^6 \equiv x^7 \equiv x \pmod{7}$$

It follows that $x^{13} + 7x + 5 \equiv x + 0 + 5 \equiv 0 \pmod{7}$, i.e. $x = -5 \equiv 2 \pmod{7}$

Consider mod 13. By the corollary of FlT, we have $x^{13} \equiv x \pmod{13}$. Thus,

$$x^{13} + 7x + 5 \equiv x + 7x + 5 \equiv 8x + 5 \equiv 0 \pmod{13}$$

By inspection, $x \equiv 1 \pmod{13}$. We remain to solve

$$\begin{cases} x \equiv 2 \pmod{7} & \text{—— (1)} \\ x \equiv 1 \pmod{13} & \text{—— (2)} \end{cases}$$

Since $\gcd(7, 13) = 1$, by CRT, there is a solution. If x satisfies (1), $x = 2 + 7y$ for some $y \in \mathbb{Z}$.

Put it into (2). We get $2 + 7y \equiv 1 \pmod{13}$, i.e. $7y \equiv -1 \pmod{13}$ —— (3)

It is equivalent to solve $7y + 13z = -1$.

CONTINUED

y	z	r	q
0	1	13	0
1	0	7	0
-1	1	6	1
2	-1	1	1
-13	7	0	6

Thus, $\gcd(7, 13) = 1$.

Since $-1 = 1 \cdot -1$, $7 \cdot -2 + 13 \cdot 1 = -1$.

Thus $y = -2$ is a solution to (3).

It follows that $x = 2 + 7(-2) = -12$ is a solution. By CRT, the complete solution is

$$x \equiv -12 \pmod{7 \cdot 13}, \text{ i.e. } x \equiv -12 \equiv 79 \pmod{91}$$

Thus $[x] = [79]$ in $\mathbb{Z}_{91}$

Theorem <u>Generalized Chinese Remainder Theorem (GCRT)</u>

Let $m_1, m_2, \ldots, m_n \in \mathbb{N}$ with $\gcd(m_i, m_j) = 1$ where $i, j \in \{1, 2, \ldots, n\}$ with $i \neq j$. Then for any $a_1, a_2, \ldots, a_n \in \mathbb{Z}$, the simultaneous congruencies

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ x &\equiv a_3 \pmod{m_3} \\ &\ldots \\ x &\equiv a_n \pmod{m_n} \end{aligned}$$

has a solution. Moreover, if $x = x_0$ is a solution, then the complete solution is

$$x \equiv x_0 \pmod{m_1 m_2 \cdots m_n}$$

Theorem : GCRT (CRT for multiple simultaneous congruencies)

CONTINUED

Ex

Solve $\begin{cases} 41x \equiv 4 \pmod{26} & \text{—— (1)} \\ 24x \equiv 6 \pmod{63} & \text{—— (2)} \end{cases}$

Note that
$$\begin{aligned} 41x &\equiv 4 \pmod{26} \\ 15x &\equiv 4 \pmod{26} \\ 15x &\equiv 30 \pmod{26} \\ x &\equiv 2 \pmod{26} \quad \text{—— (3)} \end{aligned}$$

For $24x \equiv 6 \pmod{63}$, we solve $24x + 63y = 6$.

x	y	r	q
0	1	63	0
1	0	24	0
-2	1	15	2
3	-1	9	1
-5	2	6	1
8	-3	3	1
-21	8	0	2

Thus, $\gcd(24, 63) = 3$ and
$24 \cdot 8 + 63 \cdot -3 = 3$.

Since $6 = 3 \cdot 2$, we have
$24 \cdot 16 + 63 \cdot -6 = 6$.

Thus $x = 16$ is a solution to (3).
Hence $x \equiv 16 \pmod{\frac{63}{3}}$,
i.e. $x \equiv 16 \pmod{21}$ —— (4)

It remains to solve

$$\begin{cases} x \equiv 2 \pmod{26} & \text{—— (3)} \\ x \equiv 16 \pmod{21} & \text{—— (4)} \end{cases}$$

Since $\gcd(26, 21) = 1$, by CRT, there is a solution.
If x satisfies (3), $x = 2 + 26y$ for some $y \in \mathbb{Z}$.

Put it into (4). We have $2 + 26y \equiv 16 \pmod{21}$,
i.e. $26y \equiv 14 \pmod{21}$. Hence $5y \equiv 14 \pmod{21}$.

It is equivalent to solve $5y + 21z = 14$.

x	y	r	q
0	1	21	0
1	0	5	0
-4	1	1	4
21	-5	0	5

Thus, $\gcd(5, 21) = 1$ and
$5 \cdot -4 + 21 \cdot 1 = 1$.

Since $14 = 1 \cdot 14$, we have
$5 \cdot -56 + 21 \cdot 14 = 14$

Thus $y = -56$ is a solution for (5).

Hence $x = 2 + 26 \cdot -56 = -145$ is a solution.

It follows that $x \equiv -1454 \pmod{26 \cdot 21}$,
i.e. $x \equiv -1454 \equiv 184 \pmod{546}$.